

Review retrospective:
*the mean FDT asymptote $\rightarrow -1/\lambda$, and a clean cross-library
bridge*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicFDTMeanAsymptotic.lean`: the large- t first-cumulant FDT asymptote $\partial_h \langle x \rangle_h|_0 \rightarrow -1/\lambda$, plus two definitional bridges identifying the abstract `Threepoint` Gibbs expectation/covariance at $h = 0$ with the concrete `Laplace` ones. Result: **a clean fidelity pass, no edits** — the bridges, the sign chain, and the value all check out.

1 Setting

The FDT capstone proved $\partial_h \langle x \rangle_h|_0 = -t \text{Cov}_0(x, x)$ for the anharmonic Gibbs measure. Its $t \rightarrow \infty$ asymptote needs the variance asymptotic $t \text{Cov}_0(x, x) \rightarrow 1/\lambda$ — but that asymptotic is phrased with the concrete `Laplace.gibbsCov` while the FDT identity uses the abstract `Threepoint.gibbsCov`. This file supplies the bridge and lands the asymptote.

2 What was reviewed

Three public theorems: the expectation bridge, the covariance bridge, and the headline asymptote.

3 Fidelity / soundness findings

Sound; GPT found no mismatch.

- **Bridge (expectation)**: at $h = 0$ with $A = \text{id}$, the perturbation $h \cdot A = 0$, so both `Threepoint.gibbsExp` and `Laplace.gibbsExpectation` reduce to the same normalised Lebesgue integral $\int \varphi e^{-tL} / \int e^{-tL}$ (`unfold + simp [zero_mul, add_zero]`).
- **Bridge (covariance)**: $\text{Cov} = E[\varphi\psi] - E[\varphi]E[\psi]$, so rewriting the three `gibbsExp` occurrences by the expectation bridge reduces the `Threepoint` covariance at $h = 0$ to the `Laplace` one.
- **Asymptote**: the sign chain $t \text{Cov} \rightarrow 1/\lambda \Rightarrow -(t \text{Cov}) \rightarrow -1/\lambda$ (`.neg + -(1/\lambda) = -1/\lambda`), composed with the FDT identity $(\partial_h \langle x \rangle)_0 = -t \text{Cov}$, via `.deriv` and the covariance bridge), gives $\partial_h \langle x \rangle|_0 \rightarrow -1/\lambda$ — the harmonic FDT slope.

4 Simplifications applied

None. GPT's (B) findings are all proof golf (`simp` for the two bridges, the `hlim heq`, and the final congruence); declined. No redundant hypothesis, no confident dead import/open.

5 Disagreements and open questions

None.

6 Lessons

- **A cross-library bridge at the unperturbed point is a clean `unfold+simp`.** Two formalisations of the same Gibbs object (an abstract perturbed `Threepoint.gibbsCov` and a concrete `Laplace.gibbsCov`) coincide at $h = 0$ because the perturbation term vanishes; the bridge is a one-line definitional reduction, and it is exactly what lets an asymptotic phrased in one library be composed with an identity phrased in the other. Such bridges are low-risk and high-leverage — worth a quick independent confirmation that the $h = 0$ specialisation really collapses both sides identically.

7 Follow-ups

- The laplace OneD anharmonic arc has many never-reviewed siblings (the second/third cumulant, the cumulants-from-log-partition, the general- h susceptibility, and more) — a fresh-context review arc.